

PhysicalVRF

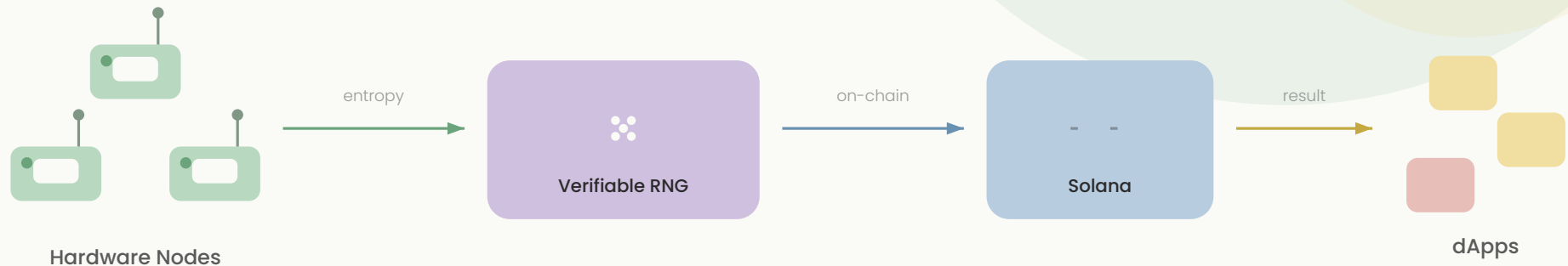
Hardware-Backed Verifiable Randomness

Solana

Hardware Entropy

Verifiable

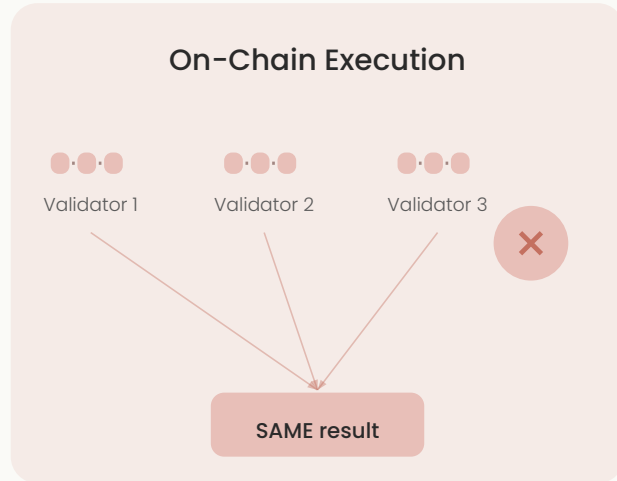
Decentralized



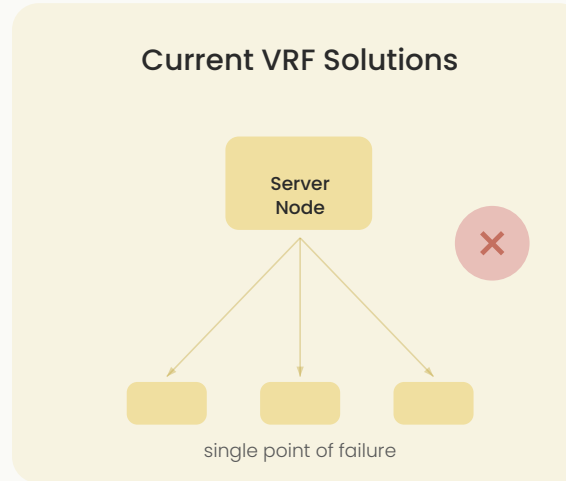
Real entropy · Real devices · Provably fair randomness

The Problem

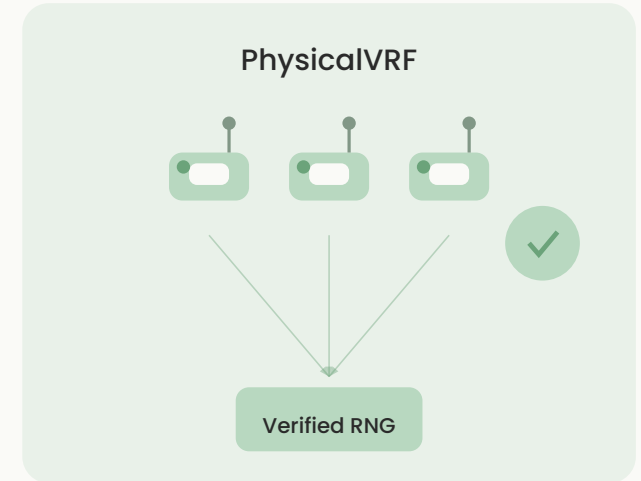
Blockchains are deterministic. They cannot produce true randomness.



Deterministic = No randomness



Centralized = Single failure

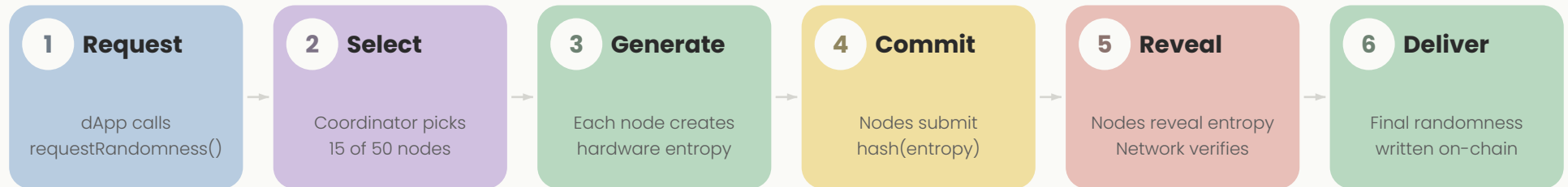


Distributed hardware = Secure

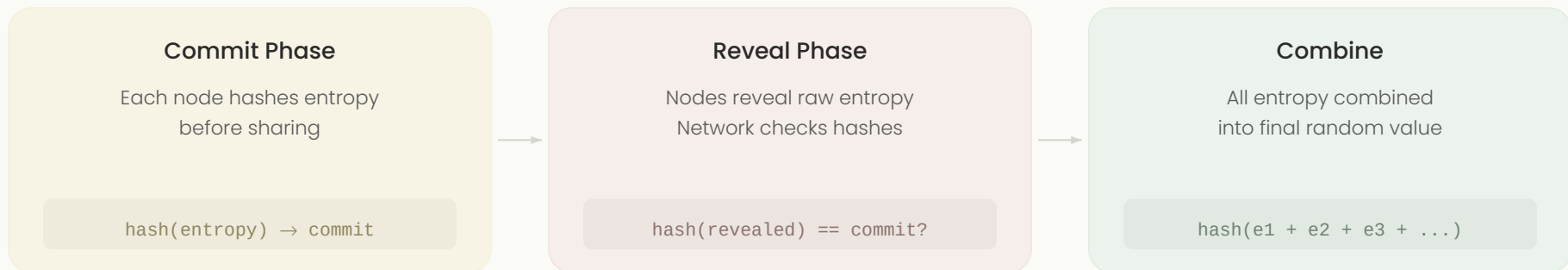
PhysicalVRF solves this with real hardware generating real entropy

How It Works

Six steps from request to verified randomness — all in under 5 seconds.

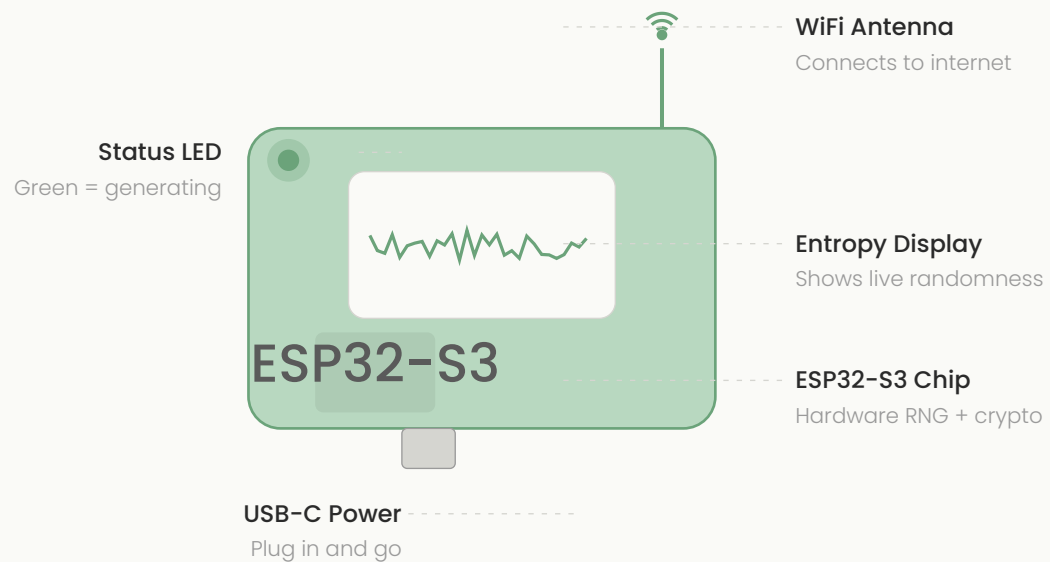


Commit-Reveal Protocol — Why It's Secure



The Hardware Node

A small plug-and-play device that generates verifiable entropy.



Entropy Sources

Hardware RNG

Clock Jitter

Environmental Noise

System Timing

Running a node is simple:

Plug in device

→ Connect WiFi

→ Stay online

→ Earn rewards

Who Needs This



Solana dApp Developers

- Games & lotteries
- NFT mint platforms
- Need provably fair RNG
- Integrate via smart contract



DeFi Protocol Teams

- Prediction markets
- Governance selection
- Fair token launches
- Tamper-proof randomness



Node Operators & HW Enthusiasts

- Run IoT / validators
- Plug-and-play device
- Earn per contribution
- Join the DePIN wave

Find them on Discord · Telegram · GitHub · X

Why Solana

Low Fees



Sub-cent per txn.
Thousands daily.

Fast Finality



2-5 second delivery.
~400ms slots.

High TPS



65,000+ transactions
per second.

Ecosystem



Games, NFTs, DeFi
already need RNG.

What PhysicalVRF Contributes Back

New Infra Primitive

Fills a missing
randomness layer

DePIN Growth

Hardware operators
join Solana

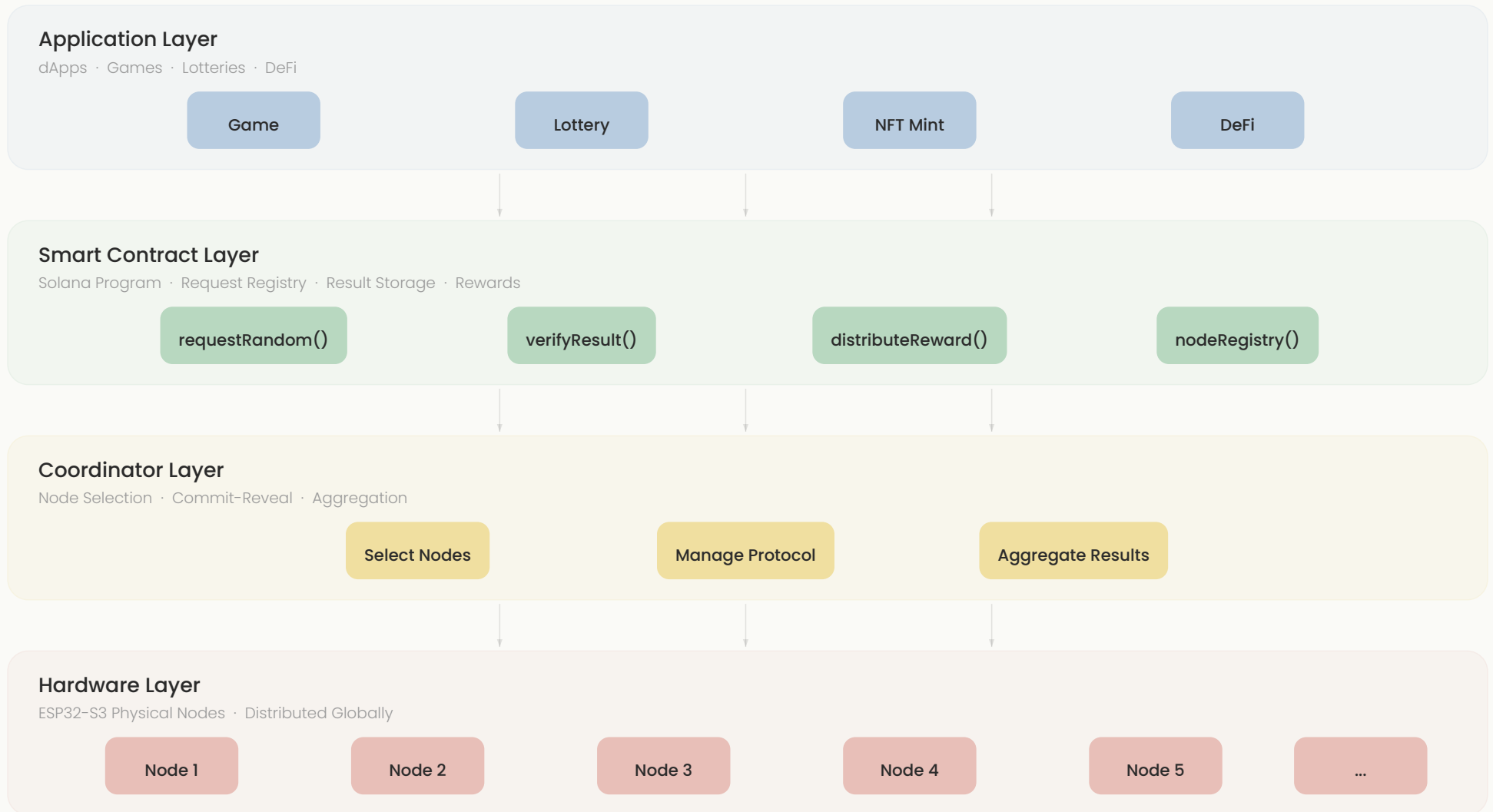
Dev Tooling

Open SDK for
fair randomness

Economic Activity

Apps pay, nodes earn
on-chain loop

System Architecture

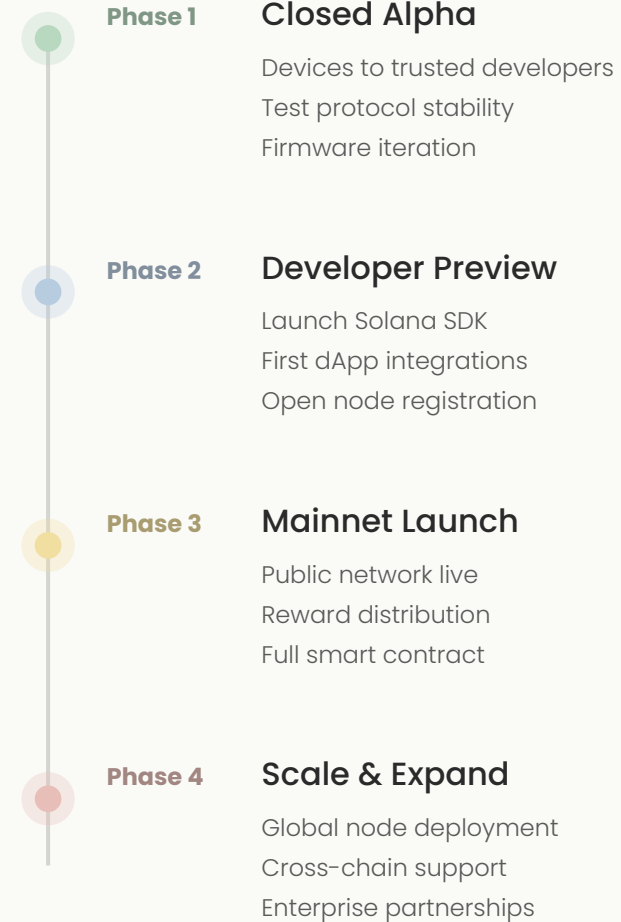


Economics & Roadmap

Economic Loop



Roadmap



In One Look



PhysicalVRF Decentralized randomness from physical hardware nodes.



The Problem Blockchains can't generate randomness. Current VRFs are centralized.



The Solution ESP32-S3 devices generate real entropy with commit-reveal protocol.



Why Hardware Physical entropy is unpredictable in ways software can never be.



Why Solana Sub-cent fees. 2-5s finality. Massive ecosystem needing RNG.



For Developers Integrate via Solana smart contract. Get verifiable random results.



For Operators Plug in a device. Connect WiFi. Earn rewards for entropy.



Security One honest node keeps the entire result secure.



Economics Apps pay per request. Nodes earn per contribution. Self-sustaining.



Vision A global physical randomness layer for the decentralized world.

Real entropy · Real devices · Provably fair randomness

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